EXP.NO-15	Career Pathways and the Future of Low-Code Development

AIM:

To implement and apply advanced performance optimization techniques—specifically Field Indexing, Subform structuring, and Deluge Query Tuning—in a Zoho Creator application to enhance execution speed, reduce response latency, and optimize database load.

THEORY:

As low-code applications scale and handle larger volumes of transactional data, inefficient data structures or poorly constructed Deluge scripts can lead to performance bottlenecks. Zoho Creator provides several built-in mechanisms and best practices to optimize application performance:

Field Indexing:

Indexing creates a database-level index on specific fields (such as Email, Unique IDs, or frequently searched criteria fields). This significantly speeds up record searches, filtering operations, and conditional lookups (fetch / zoho.creatoriq queries) by avoiding full table scans.

Efficient Subform Usage:

Subforms allow parent-child relationships within a single form view (e.g., an Order form containing multiple Order Item subform rows). Grouping related transactional lines reduces the need for multiple independent lookup forms, streamlines data entry, and optimizes relational queries when fetched via Deluge.

Deluge Query Tuning & Best Practices:

Writing optimized Deluge scripts by minimizing database read/write calls within loops, using precise criteria filters, and limiting fetched records ([Criteria].sort(ID, false).range(1, 50)). This reduces API call consumption, avoids execution time-outs, and ensures smooth user interface responsiveness.

PROCEDURE:

1. Open the Zoho Creator Developer workspace and select Patient Appointment & Scheduling Management

2. Open Appointment_Booking_Form, inspect frequently searched fields, and review available indexing settings. Do not impose unique email constraints solely for performance.

3. Verify the existing Appointment Services subform. Keep essential columns such as Service Name and limit the number of entries to maintain a manageable form.

4. Navigate to Workflow > Functions and open the Optimized_Appointment_Query Deluge function.

5. Inspect the query filtering by Preferred_Appointment_Date and limiting the result set to a maximum of 50 upcoming appointment records.
appointments = Appointment_Booking_Form [Preferred_Appointment_Date >= zoho.currentdate] sort by Preferred_Appointment_Date range from 1 to 50;

6. Avoid broad queries and unnecessary fetch or update operations inside loops; use precise criteria and process only the needed records.

7. Use the script executor, relevant execution logs, and repeatable test conditions to observe record counts and run times. Compare before and after measurements only where measurements exist.

RESULT:

The Patient Appointment & Scheduling Management project demonstrates subform structuring through Appointment Services and targeted record retrieval through the Optimized_Appointment_Query Deluge function. Common indexing considerations and monitoring practices were reviewed. No database index creation or quantified performance improvement is asserted without direct configuration or measured execution results.